μPD27128
16,384 x 8-BIT
NMOS UV/OTP EPROM

Revision 1

Description

The μPD27128 is a 131,072-bit (16,384 × 8) electrically programmable read-only memory (EPROM). It operates from a single +5V supply making it ideal for microprocessor applications. It features an output enable control and offers a standby mode with reduction in power consumption.

A distinctive feature of the μPD27128 is a separate output enable control ($\overline{OE}$) in addition to the chip enable control ($\overline{CE}$). The $\overline{OE}$ control eliminates bus contention in multiple-bus microprocessor systems. The μPD27128 features conventional, simple one-pulse programming controlled by TTL-level signals as well as a high-speed programming mode. Total programming time for all 131,072 bits is 820 seconds for the conventional mode, and typically 120 seconds for the high-speed mode.

The μPD27128 is available in a cerdip package as an ultraviolet (UV), erasable EPROM, or in a plastic package as a one-time-programmable (OTP), non-erasable EPROM.

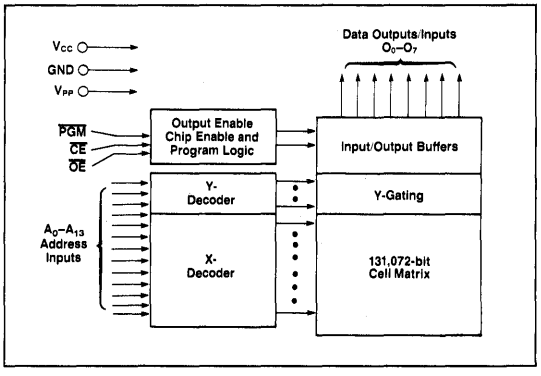
Features

- ☐ Ultraviolet erasable and electrically programmable
- ☐ Access time—200ns max
- ☐ Low power dissipation: 100mA max active current
  25mA max standby current
- ☐ High-speed programming mode
  (typical program time 120s)
- ☐ Programmable with single pulse
  (total program time 820s)
- ☐ Industry standard pinout (JEDEC approved)
- ☐ 4 performance ranges

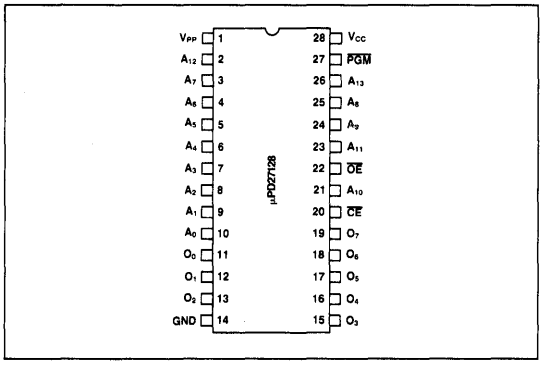
Device	Max Access Time	Max V_{CC} Supply Current	
		Active	Standby
μPD27128-2	200ns	100mA	25mA
μPD27128①	250ns	100mA	25mA
μPD27128-3①	300ns	100mA	25mA
μPD27128-4①	450ns	100mA	25mA

Note: ① Available as either UV or OTP.

Block Diagram



Pin Configuration



Pin Identification

A_0–A_{13}	Addresses
$\overline{OE}$	Output Enable
O_0–O_7	Data Outputs
$\overline{CE}$	Chip Enable
$\overline{PGM}$	Program

Mode Selection

Mode \ Pins	$\overline{CE}$ (20)	$\overline{OE}$ (22)	$\overline{PGM}$ (27)	V_{PP} (1)	V_{CC} (28)	Outputs (11–13, 15–19)
Read	V_{IL}	V_{IL}	V_{IH}	V_{CC}	V_{CC}	D_{OUT}
Standby	V_{IH}	X	X	V_{CC}	V_{CC}	High Z
Program	V_{IL}	V_{IH}	V_{IL}	V_{PP}	V_{CC}	D_{IN}
Program Verify	V_{IL}	V_{IL}	V_{IH}	V_{PP}	V_{CC}	D_{OUT}
Program Inhibit	V_{IH}	X	X	V_{PP}	V_{CC}	High Z
High Speed Programming	V_{IL}	V_{IH}	V_{IL}	V_{PP}	V_{CC}	D_{IN}

Note: X can be either V_{IL} or V_{IH}.

Absolute Maximum Ratings*

Operating Temperature	−10°C to +80°C
Storage Temperature	−65°C to +125°C
Output Voltage	−0.6V to 7.0V
Input Voltage	−0.6V to 7.0V
Supply Voltage V_{CC}	−0.6V to 7.0V
Supply Voltage V_{PP}	−0.6V to +22V

*COMMENT: Exposing the device to stresses above those listed in Absolute Maximum Ratings could cause permanent damage. The device is not meant to be operated under conditions outside the limits described in the operational sections of this specification. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Capacitance

T_A = 25°C; f = 1MHz

Parameter	Symbol	Limits Min	Limits Typ	Limits Max	Unit	Test Conditions
Input Capacitance	C_{IN}		4	8	pF	V_{IN} = 0V
Output Capacitance	C_{OUT}		8	14	pF	V_{OUT} = 0V

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DC Characteristics

Read Mode and Standby Mode

T_A = 0°C to + 70°C; V_{CC} = +5V ± 5%; V_{PP} = V_{CC}

Parameter		Symbol	Limits Min	Typ	Max	Unit	Test Conditions
Output High Voltage		V_{OH}	2.4			V	I_{OH} = −400μA
Output Low Voltage		V_{OL}			0.45	V	I_{OL} = 2.1mA
Input High Voltage		V_{IH}	2.0		V_{CC} + 1	V	
Input Low Voltage		V_{IL}	−0.1		0.8	V	
Output Leakage Current		I_{LO}			10	μA	V_{OUT} = 5.25V
Input Leakage Current		I_{LI}			10	μA	V_{IN} = 5.25V
V_{CC} Current	Standby	I_{CC1}			25	mA	$\overline{CE}$ = V_{IH}
	Active	I_{CC2}		60	100	mA	$\overline{OE}$ = $\overline{CE}$ = V_{IL}
V_{PP} Current		I_{PP1}			15	mA	V_{PP} = 5.25V

Program, Program Verify, and Program Inhibit Modes

T_A = 25°C ± 5°C; V_{CC} ① = +5V ± 5%; V_{PP} = + 21V ± 0.5V

Parameter		Symbol	Limits Min	Typ	Max	Unit	Test Conditions
Input High Voltage		V_{IH}	2.0		V_{CC} + 1	V	
Input Low Voltage		V_{IL}	−0.1		0.8	V	
Input Leakage Current		I_{LI}			10	μA	V_{IN} = V_{IL} or V_{IH}
Output High Voltage		V_{OH}	2.4			V	I_{OH} = −400μA
Output Low Voltage		V_{OL}			0.45	V	I_{OL} = 2.1mA
V_{CC} Current	Program Inhibit	I_{CC1}			25mA		$\overline{CE}$ = V_{IH}
	Program Verify	I_{CC2}			100mA		
V_{PP} Current	Program	I_{PP2}			30mA		$\overline{CE}$ = $\overline{PGM}$ = V_{IL}
	Program Verify	I_{PP3}			15mA		$\overline{CE}$ = V_{IL}, $\overline{PGM}$ = V_{IH}
	Program Inhibit	I_{PP4}			15mA		$\overline{CE}$ = V_{IH}

Note: ① V_{CC} = 6V ± 0.25V for high-speed programming.

AC Characteristics

Read Mode and Standby Mode

T_A = 0°C to +70°C; V_{CC} = +5V ± 5%; V_{PP} = V_{CC}

Parameter	Symbol	Limits 27128-2 Min	27128-2 Max	27128① Min	27128① Max	27128-3① Min	27128-3① Max	27128-4① Min	27128-4① Max	Unit	Test Conditions
Address to Output Delay	t_{ACC}		200		250		300		450	ns	$\overline{CE}$ = $\overline{OE}$ = V_{IL}
$\overline{CE}$ to Output Delay	t_{CE}		200		250		300		450	ns	$\overline{OE}$ = V_{IL}
Output Enable to Output Delay	t_{OE}		75		100		120		150	ns	$\overline{CE}$ = V_{IL}
Output Enable High to Output Delay	t_{DF}	0	60	0	85	0	105	0	130	ns	$\overline{CE}$ = V_{IL}
Address to Output Hold Time	t_{OH}	0		0		0		0		ns	$\overline{CE}$ = $\overline{OE}$ = V_{IL}

Note: ① Available as either UV or OTP.

Test Conditions—
Output Load: See Fig. 1.
Input Rise and Fall Times: 20ns
Input Pulse Levels: 0.45V to 2.4V
Timing Measurement Reference Levels:
Inputs: 0.8V and 2.0V
Outputs: 0.8V and 2.0V

AC Characteristics (Cont.)

Program, Program Verify, and Program Inhibit Modes

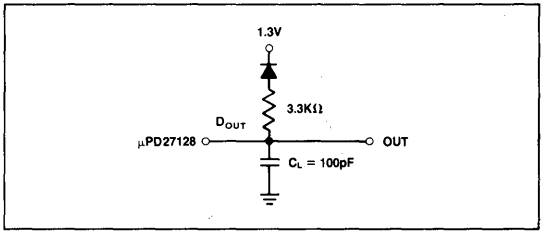
T_A = 25°C ± 5°C; V_{CC} = +5V ± 5%; V_{PP} = +21V ± 0.5V

Parameter	Symbol	Limits Min	Typ	Max	Unit	Test Conditions
Address Setup Time	t_{AS}	2			μs	Input Pulse Levels = 0.45V to 2.4V Input Timing Reference Level = 0.8V and 2.0V Output Timing Reference Level = 0.8V and 2V Input Rise and Fall Times: 20ns
$\overline{OE}$ Setup Time	t_{OES}	2			μs	
Data Setup Time	t_{DS}	2			μs	
Address Hold Time	t_{AH}	0			μs	
$\overline{CE}$ Setup Time	t_{CES}	2			μs	
Data Hold Time	t_{DH}	2			μs	
Chip Enable to Output Float Delay	t_{DF}	0		130	ns	
Data Valid from $\overline{OE}$	t_{OE}			150	ns	
Program Pulse Width①	t_{PW}	45	50	55	ms	
V_{PP} Setup Time	t_{VS}	2			μs	

Note: ①V_{CC} = 6V ± 0.25V and t_{PW} = 1 ms ± 5% for high-speed programming.

Test Conditions—
Input Pulse Levels = 0.45V to 2.4V
Input Timing Reference Level = 0.8V and 2.0V
Output Timing Reference Level = 0.8V and 2V
Input Rise and Fall Times: 20ns

Figure 1. Loading Conditions Test Circuit



Function

The μPD27128 operates from a single +5V power supply making it ideal for microprocessor applications.

The μPD27128 features a standby mode which reduces the power dissipation.

Operation

The six operation modes of the μPD27128 are listed in Table 1. In the read mode the only power supply required is a +5V supply. During programming all inputs are TTL levels except for V_{PP} which rises from V_{CC} level to 21V.

Read Mode

When $\overline{CE}$ and $\overline{OE}$ are at a low (0) level, Read is set and data is available at the outputs after t_{OE} from the falling edge of $\overline{OE}$ and t_{ACC} after setting the address.

Standby Mode

The μPD27128 is placed in a standby mode with the application of a high (1) level TTL signal to the $\overline{CE}$ input. In this mode the outputs are in a high impedance state, independent of the $\overline{OE}$ input. The active power dissipation is also reduced.

Programming Modes

The μPD27128 can be programmed in two ways: (1) conventional programming mode, and (2) high-speed programming mode. In the conventional mode, basically a 50ms $\overline{PGM}$ pulse is applied to each bit location. The high-speed

programming mode is similar to the Intelligent Programming Algorithm™, in which up to fifteen 1ms $\overline{PGM}$ pulses are applied to each bit location, followed by an additional 4ms $\overline{PGM}$ pulse for each number of 1ms pulses applied before. The high-speed programming mode reduces the programming time to 120s typical.

Conventional Programming Mode

Programming begins with erasing all data and consequently having all bits in the high (1) level state. Data is then entered by programming a low (0) level TTL signal into the chosen bit location.

The μPD27128 is placed in the programming mode by applying a low (0) level TTL signal to the $\overline{CE}$ and $\overline{PGM}$ with V_{PP} at +21V. The data to be programmed is applied to the output pins in 8-bit parallel form at TTL levels.

Any location can be programmed at any time, either individually, sequentially, or at random.

When multiple μPD27128s are connected in parallel except for $\overline{CE}$, individual μPD27128s can be programmed by applying a low (0) level TTL pulse to the $\overline{PGM}$ input of the desired μPD27128 to be programmed.

Programming of multiple μPD27128s in parallel with the same data is easily accomplished. All the like inputs are tied together and programmed by applying a low (0) level TTL pulse to the $\overline{PGM}$ inputs.

High-speed Programming Mode

In this mode, programming begins by addressing the first location, and valid data appearing at the eight output pins (a low level TTL signal, 0, into the chosen bit location).

V_{CC} is then raised to 6V ± 0.25V followed by V_{PP} raised to 21V ± 0.5V. A $\overline{PGM}$ pulse of 1ms ± 5% is then applied in the same manner as described in the program mode timing diagram. The bit is then verified and a program/no program decision is made. If the bit is not programmed, another 1ms $\overline{PGM}$ pulse is applied, to a maximum of fifteen times. If the bit gets programmed within fifteen efforts, another pulse of 4ms for each effort is applied and the next address is applied. If the bit does not get programmed in fifteen 1ms efforts, another $\overline{PGM}$ pulse of 60ms is applied and the bit verified. If the bit is not programmed at this stage, the device would be rejected as a program failure. If the bit is programmed, the next address is applied until all addresses are complete.

At this stage, V_{CC} and V_{PP} pins are lowered to 5V ± 5% and all bytes are then verified again for programming. This algorithm is compatible with that of the μPD2764.

Program Verify Mode

A verify should be performed on the programmed bits to determine that the data was correctly programmed. The program verify can be performed with $\overline{CE}$ and $\overline{OE}$ at low (0) levels and $\overline{PGM}$ at a high (1) level.

Programming Inhibit Mode

Programming multiple μPD27128s in parallel with different data is easier with the program inhibit mode. Except for $\overline{CE}$ (or $\overline{PGM}$), all like inputs (including $\overline{OE}$) of the parallel μPD27128s may be common. Programming is accomplished by applying a low (0) TTL-level program pulse to the $\overline{CE}$ (or $\overline{PGM}$) input with V_{PP} at +21V. A high (1) level applied to the $\overline{CE}$ (or $\overline{PGM}$) of the other μPD27128s will inhibit it from being programmed.

Output Disable

The data outputs of two or more μPD27128s may be wire-ORed together to the same data bus. In order to prevent bus contention problems between devices, all but the selected μPD27128 should be disabled by raising the $\overline{CE}$ input to a TTL high. $\overline{OE}$ input should be made common to all devices and connected to the read line from the system control bus. These connections offer the lowest average power consumption.

Erasure Mode

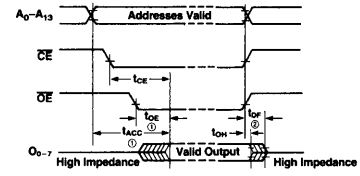
Erasure of the μPD27128 programmed data can be attained when exposed to light with wavelengths shorter than approximately 4,000 Angstroms (Å). It should be noted that constant exposure to direct sunlight or room level fluorescent lighting could erase the μPD27128. Consequently, if the μPD27128 is to be exposed to these types of lighting conditions for long periods of time, its window should be masked to prevent unintentional erasure. Opaque labels are supplied with every device.

The recommended erasure procedure for the μPD27128 is exposure to ultraviolet light with wavelengths of 2,537 Angstroms (Å). The integrated dose (i.e., UV intensity × exposure time) for erasure should be not less than 15W-sec/cm^2. The erasure time is approximately 15 to 20 minutes using an ultraviolet lamp of 12,000μW/cm^2 power rating.

During erasure, the μPD27128 should be placed within 1 inch of the lamp tubes. If the lamps have filters on the tubes, the filters should be removed before erasure.

Timing Waveforms

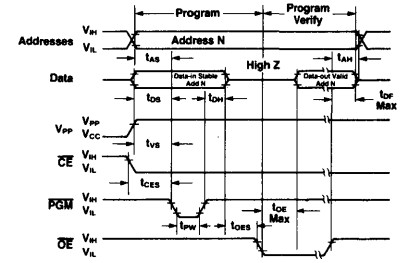
Read Mode



Notes: ① $\overline{OE}$ may be delayed up to $t_{ACC} - t_{OE}$ after the falling edge of $\overline{CE}$ for read mode without impact on t_{ACC}.

② t_{DF} is specified from $\overline{OE}$ or $\overline{CE}$, whichever occurs first.

Program Mode



μPD27256
32,768 x 8-BIT
NMOS UV EPROM

Revision 2

Description

The μPD27256 is a 262,144-bit electrically programmable read-only memory utilizing NMOS double-polysilicon technology. The device is organized as 32K words by 8 bits and operates from a single +5V ±5% power supply. All inputs and outputs are TTL-compatible. The μPD27256 has single location programming, three-state outputs and is pin-compatible with the 27C256 EPROM. It is available as a 28-pin DIP.

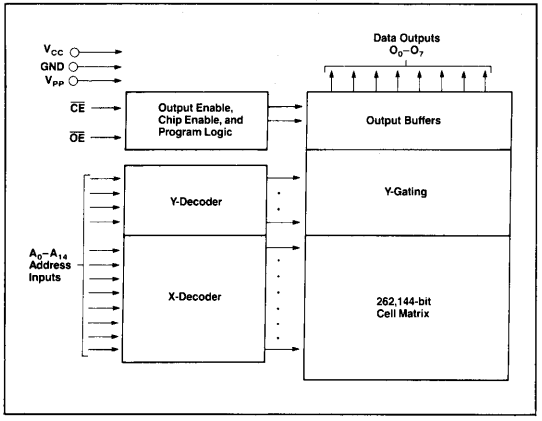
The μPD27256 is available in a cerdip package with a quartz window as an ultraviolet (UV), erasable EPROM.

Features

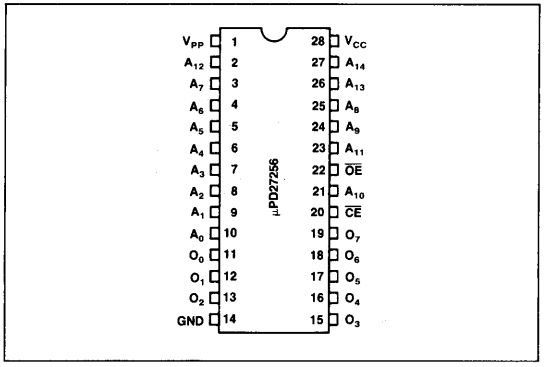
- ☐ 32K by 8 organization
- ☐ Ultraviolet erasable and electrically programmable
- ☐ Access time — 200ns max
- ☐ Single location programming
- ☐ High-speed programming mode
- ☐ Low power dissipation: 100mA max (active)
  25mA max (standby)
- ☐ Input/output TTL-compatible for reading and programming
- ☐ Single +5V ±5% power supply
- ☐ Three-state outputs
- ☐ Pin-compatible with μPD27C256 EPROM
- ☐ NMOS double-polysilicon technology
- ☐ 28-pin DIP
- ☐ 3 performance ranges:

Device	Access Time	Power Supply	
		Active	Standby
μPD27256	250ns	100mA	25mA
μPD27256-3	300ns	100mA	25mA

Block Diagram



Pin Configuration



Pin Identification

Pin No.	Symbol	Description
1	V_{PP}	Program Voltage
2–10, 21, 23–27	A_0–A_{14}	Address Inputs
11–13, 15–19	O_0–O_7	Data Outputs
14	GND	Ground
20	$\overline{CE}$	Chip Enable
22	$\overline{OE}$	Output Enable
28	V_{CC}	+5V ±5% Power Supply

Mode Selection

Mode \ Pins	$\overline{CE}$ (20)	$\overline{OE}$ (22)	V_{PP} (1)	V_{CC} (28)	Outputs (11–13, 15–19)
Read	V_{IL}	V_{IL}	V_{CC}	V_{CC}	D_{OUT}
Standby	V_{IH}	X	V_{CC}	V_{CC}	High-Z
Program	V_{IL}	V_{IH}	V_{PP}	V_{CC}	D_{IN}
Program Verify	V_{IL}	V_{IL}	V_{PP}	V_{CC}	D_{OUT}
Program Inhibit	V_{IH}	X	V_{PP}	V_{CC}	High-Z

Note: X can be either V_{IL} or V_{IH}.

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Absolute Maximum Ratings*

Operating Temperature, T_{OPR}	−10°C to +80°C
Storage Temperature, T_{STG}	−65°C to +125°C
Output Voltage, V_{OH}	−0.6V to V_{CC} +6.5V
Input Voltage, V_{IH}	−0.6V to V_{CC} +6.5V
Supply Voltage, V_{CC}	−0.6V to +7V
Supply Voltage, V_{PP}	−0.6V to +22V

*COMMENT: Exposing the device to stresses above those listed in Absolute Maximum Ratings could cause permanent damage. The device is not meant to be operated under conditions outside the limits described in the operational sections of this specification. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Capacitance ①

T_A = 25°C; f = 1MHz

Parameter	Symbol	Limits Min	Typ	Max	Unit	Test Conditions
Input Capacitance	C_{IN}		4	8	pF	V_{IN} = 0V
Output Capacitance	C_{OUT}		8	14	pF	V_{OUT} = 0V

Note: ① This parameter is sampled periodically.

DC Characteristics

T_A = 0°C to +70°C; V_{CC} = +5V ±5%; V_{PP} = V_{CC}
Read and Standby Modes

Parameter	Symbol	Limits Min	Typ	Max	Unit	Test Conditions
Output High Voltage	V_{OH}	2.4			V	I_{OH} = −400μA
Output Low Voltage	V_{OL}			0.45	V	I_{OL} = 2.1mA
Input High Voltage	V_{IH}	2.0		V_{CC}+1	V	
Input Low Voltage	V_{IL}	−0.1		0.8	V	
Output Leakage Current	I_{LO}			10	μA	$\overline{OE}$ = V_{IH}; V_{OUT} = 0 to V_{CC}
Input Leakage Current	I_{LI}			10	μA	V_{IN} = 0 to V_{CC}
V_{CC} Current	I_{CC1}			25	mA	$\overline{CE}$ = V_{IH} (standby)
	I_{CC2}			100	mA	$\overline{CE}$ = V_{IL} (active)
V_{PP} Current	I_{PP1}			15	mA	V_{PP} = 5.25V

T_A = 25°C ± 5°C; V_{CC} = 6V ± 0.25V; V_{PP} = +21V ± 0.5V
Program, Program Verify, and Program Inhibit Modes

Parameter	Symbol	Limits Min	Typ	Max	Unit	Test Conditions
Input High Voltage	V_{IH}	2.0		V_{CC}+1	V	
Input Low Voltage	V_{IL}	−0.1		0.8	V	
Input Leakage Current	I_{LI}			10	μA	V_{IN} = V_{IL} or V_{IH}
Output High Voltage	V_{OH}	2.4			V	V_{OH} = −400μA
Output Low Voltage	V_{OL}			0.45	V	I_{OL} = 2.1mA
V_{CC} Current	I_{CC2}			100	mA	
V_{PP} Current	I_{PP2}			30	mA	$\overline{CE}$ = V_{IL}; $\overline{OE}$ = V_{IH}

AC Characteristics

T_A = 0°C to +70°C; V_{CC} = +5V ± 5%; V_{PP} = V_{CC}
Read and Standby Modes

Parameter	Symbol	Limits 27256 Min	27256 Max	27256-3 Min	27256-3 Max	Unit	Test Conditions ①
Address to Output Delay	t_{ACC}		250		300	ns	$\overline{CE}$ = $\overline{OE}$ = V_{IL}
CE to Output Delay	t_{CE}		250		300	ns	$\overline{CE}$ = V_{IL}
Output Enable to Output Delay	t_{OE}		100		120	ns	$\overline{CE}$ = V_{IL}
Output Enable High to Output Float	t_{DF}	0	85		105	ns	$\overline{CE}$ = V_{IL}
Address to Output Hold	t_{OH}	0		0		ns	$\overline{CE}$ = V_{IL}

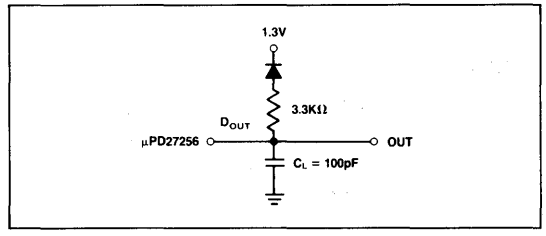
Notes: ① Output load: see Figure 1.
Input rise and fall times: 20ns.
Input pulse levels: 0.45V to 2.4V.
Input and output timing measurement reference levels: 0.8V and 2.0V.

T_A = 25°C ± 5°C; V_{CC} = +6V ± 0.25V; V_{PP} = 21V ± 0.5V
Program, Program Verify, and Program Inhibit Modes ① ②

Parameter	Symbol	Limits Min	Typ	Max	Unit	Test Conditions
Address Set-up Time	t_{AS}	2			μs	
Data Set-up Time	t_{DS}	2			μs	
Data Hold Time	t_{DH}	2			μs	
Address Hold Time	t_{AH}	2			μs	Input pulse levels = 0.45V to 2.4V
Chip Enable to Output Float Delay	t_{DF}			130	ns	Input and output timing reference levels = 0.8V and 2.0V
V_{CC} = Set-up Time	t_{VS}	2			μs	
Program Pulse Width	t_{PW}	0.95	1	1.05	ms	
$\overline{CE}$ Set-up Time	t_{CES}	2			μs	Input rise and fall times = 20ns
$\overline{OE}$ Set-up Time	t_{OES}	2			μs	
$\overline{OE}$ Hold Time ③	t_{OEH}	2			μs	
$\overline{OE}$ Recovery Time ③	t_{OR}	2			μs	
$\overline{CE}$ to Output Valid ③	t_{DV}			1	μs	

Notes: ① V_{IH} (min) and V_{IL} (max) are reference levels for measuring timing of input signals. Transition times are measured between V_{IH} and V_{IL}.
② V_{CC} must be applied simultaneously or before V_{PP} and removed after V_{PP}.
③ These parameters are sampled periodically.

Figure 1. Loading Conditions Test Circuit



Programming Operation

High Speed Programming Mode

Begin programming by erasing all data; this places all bits in the high level (1) state. Enter data by programming a low level (0) TTL signal into the chosen bit location.

Address the first location and apply valid data at the 8 output pins. Raise V_{CC} to +6 V ±.25 V; then raise V_{PP} to +21 V ±.5 V. Apply a 1 ms (±5%) program pulse to $\overline{CE}$ as shown in the programming mode timing waveform. The bit is verified and the program/no-program decision is made. If the bit is not programmed, apply another 1 ms pulse to $\overline{CE}$ up to a maximum of 20 times. If the bit is programmed within 20 tries, apply an additional overprogram pulse of (1 x number of tries) ms and input the next address. If the bit is not programmed in 20 tries, reject the device as a program failure.

After all bits are programmed, lower both V_{CC} and V_{PP} to +5 V ±5% and verify all data again.

Programming Inhibit Mode

Use the programing inhibit mode to program multiple μPD27256s connected in parallel. All like inputs (except $\overline{CE}$, but including $\overline{OE}$) may be common. Program individual devices by applying a low level (0) TTL pulse to the $\overline{CE}$ input of the μPD27256 to be programmed. Applying a high level (1) to the $\overline{CE}$ input of the other devices prevents them from being programmed.

Program Verify Mode

Perform verification on the programmed data to determine that the data was correctly programmed. The program verification can be performed with the $\overline{CE}$ and $\overline{OE}$ at low levels (0).

Erasure

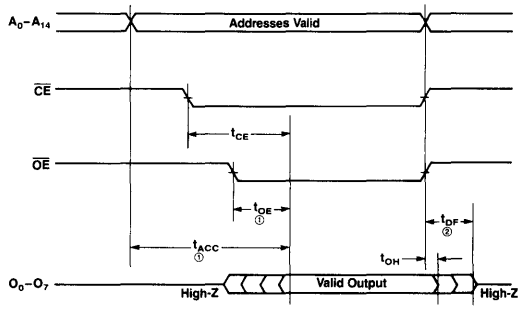
Erase data on the μPD27256 by exposing it to light with a wavelength shorter than 400 nm. Exposure to direct sunlight or fluorescent light could also erase the data. Consequently, mask the window to prevent unintentional erasure by ultraviolet rays.

Data is typically erased by 254 nm ultraviolet rays. A lighting level of 15 W-sec/cm^2 (min) is required to completely erase written data (ultraviolet ray intensity x exposure time.)

An ultraviolet lamp rated at 12,000 μW/cm^2 takes approximately 15 to 20 minutes to complete erasure. Place the μPD27256 within 2.5 cm of the lamp tubes. Remove any filter on the lamp.

Timing Waveforms

Read Mode



Notes: ① $\overline{OE}$ may be delayed up to $t_{ACC} - t_{OE}$ after the falling edge of $\overline{CE}$ for read mode without impact on t_{ACC}.
② t_{DF} is specified from $\overline{OE}$ or $\overline{CE}$, whichever occurs first.

Program Mode

